

Shell Eco Marathon 2015 - Asia Team HammerHead Arc Sponsor and Partnership Proposal



GHULAM ISHAQ KHAN INSTITUTE



ABOUT SHELL ECO MARATHON

As mankind reaches a deadlock between shielding nature and our own technological needs, someone must strive to pave the way for a mutually beneficial solution. Shell Eco-marathon provides an international platform for pioneers striving for an answer for this conundrum.

Shell Eco-marathon has set the benchmark for innovation in recent years galvanizing university students to contest globally to make ultra-efficient vehicles. The event sparks debate about the future of mobility and inspires young engineers to push the boundaries of fuel efficiency for a bright future.



GHULAM ISHAQ KHAN INSTITUTE OF ENGINEERING SCIENCES & TECHNOLOGY

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) inspires to be the focal point of engineering innovation and excellence as one of the prestigious engineering institutions in Pakistan.

For the last five years, GIKI has made the country proud in Shell Eco-marathon by gaining recognition in different domains pertaining to design, manufacturing, and communication. This year, Team Hammerhead Arc aims to uphold the legacy of our predecessors.

Previous accomplishments:

- February 2014, Team Toyota Hammerhead Arc finished at 6th position in battery-electric category. Team impressed the Shell Asia Manager and Pakistan's Ambassador to Philippines visited and commended team's innovative creation.
- 2012 Malaysia, Team Hammerhead achieved record mileage of 270km/liter, three times the mileage received by any Pakistani team before and won praise from Shell supervisors for its aerodynamic design and light weight.
- 2010 & 2011 Malaysia, Team made a national record of 90km/liter. Team Aqua I & II qualified for design and safety awards.
- 2009 Germany, Team Exodus (GIKI) won an honorary mention in the communication and marketing award.



THE TEAM

Team HammerHead ARC is comprised of undergraduates from different faculties at GIKI. It is completely organised and run by student members with guidance from advisers only when required. Selected through a competitive induction process, each member brings a unique skill set to the team. All the project's domains, from manufacturing the car to finances and from ensuring safety to participation in off-track awards, require serious commitment.

Team has access to facilities on campus which include dedicated lab area, machine workshop, and many types of specialized equipment.

PROJECT ADVISOR

Prof. Dr. Javed Ahmad Chattha

Professor, Faculty of Mechanical Engineering, GIK Institute M.Engg. at Asian Institute of Technology (AIT), Thailand
Ph.D. , Birmingham University, England, UK
Member of PEC, ASME, AIAA and ISES.

PAST WORK EXPERIENCE

Aisha Murtaza

Summer Intern - Coca Cola Beverages Pakistan Limited
Member - Team AvroCar (International DBFC Team)

Ali Yousuf

Summer Research Intern - EPFL, Switzerland
Webmaster - GIKI WebTeam
Best Engineering Design Award - NERC EME 2013

Ashaa Naveed

Exchange Participant - South Korea & Romania (AIESEC)
President - IEEE Women in Engineering, GIKI
International Leader of Tomorrow Award Scholarship - UBC

Kazi Sher Ahmed

Testing and Publication Lead - Team HammerHead (SEM 2014)
Summer Research Intern - MARC, Koc University, Turkey
Managing Editor - Science Aurora Magazine

Malik Abbas

Best Engineering Design Award - NERC EME 2013
Summer Intern - Mentor Graphics

Talha Khan

Intern - Cloud9 Networks, Dubai
Summer Intern - Lotte Chemicals
Best Engineering Design Award - NERC EME 2013

Zaire Hasan

Testing and Social Media Lead - Team HammerHead (SEM 2014)
President - AIAA GIKI Chapter
Summer Intern - K-Electric

ABOUT THE CAR

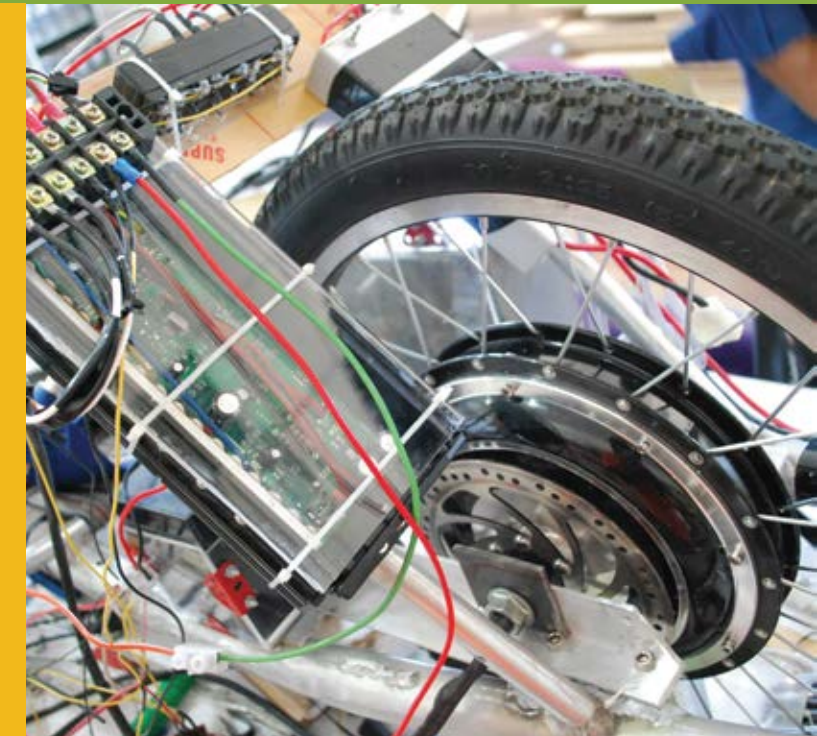
To achieve this year's targets, the team will be developing a number of innovative ideas which include:

- + Custom-made motor controller linking motors, brakes, and throttle.
- + Construction of new light-weight fiber-glass body to reduce weight.
- + Redesign of chassis for weight reduction
- + Implementation of an electronic system which includes dynamic data acquisition, power management module, maximum powerpoint tracking module, intelligent battery management system.
- + New DC motor with highest torque per ampere characteristic.
- + Highest quality Lithium-ion battery.

All of these developments require valuable time, labor, technical support and funding, which can be achieved through forming a partnership with Team HammerHead ARC.

The current main features of Team HammerHead ARC's prototype car make it possible for the team to give other teams a run for their money at Shell Eco Marathon Asia 2015. These are as follows:

- + Passed safety and technical inspection flawlessly.
- + Weight 39 Kg; considered to be one of the lightest in SEM Asia 2014.
- + Low rolling resistance Michelin tires approved by Shell.
- + Very low drag-coefficient.
- + Two-way joystick steering control which greatly enhances visibility from inside the car.



ESTIMATED COST BREAKDOWN

All prices are in Pakistani Rupees

DC Motors
30,000

Li-ion Battery
45,000

Chassis/Shell Restoration
60,000

Motor Controller
30,000

Travelling Cost	800,000
Car freight	240,000
Accommodation	160,000
Office costs	15,000
Apparel	20,000
Driver Gear	20,000

Michelin E-bike Tyres
40,000

Total Cost
1,460,000



SPONSORSHIP BENEFITS

Team HammerHead ARC welcomes all types of sponsorship including pure financial support, in-kind sponsorship, and technical assistance.

Benefits not only include placing a logo on the car but also an opportunity to be directly associated with some of the most talented young engineers in the country, this gives you an exclusive opportunity to assess potential recruits. Many of our previous sponsors have benefitted from this mutual interaction.

We highly encourage sponsors to visit us whenever possible both at the university and on track. All of this helps to generate great team spirit.

This association presents you a chance to reach your nationwide and international target audience through car appearances on event, campus and on media. Our team has previously been featured on GEO TV as well as in articles in The News, Dawn, Express Tribune, Science Aurora magazine, and GIKI's website. Not forgetting the event itself where photos of cars adorned with sponsor logos are included in Shell publications.

All donations are 100% tax deductible.

KIND NOTE:

It must be brought to the attention of our potential sponsors that many student team endeavors have not displayed their true potential in international events and competitions, mostly because of delays in fulfillment of commitments.

Therefore it is kindly requested that any firm or institution willing to commit must honor their commitments in time so that we can achieve our goals according to our project schedule.

SPONSORSHIP PACKAGES

	FRIENDS OF HAMMERHEAD ARC	BRONZE	SILVER	GOLD	TITLE
SPONSORSHIP AMOUNT*	PKR. 30,000	PKR. 150,000	PKR. 300,000	PKR. 750,000	PKR.1,000,000
Logo on car and event pit flex standee	TEXT	SMALL (Car: 100 sq. cm)	MEDIUM (Car: 200 sq. cm)	LARGE (Car: 300 sq. cm)	X-LARGE
Recognition on team website, literature, and social media	TEXT	SMALL	MEDIUM	LARGE	SPECIAL
Invitation to unveiling event	✓	✓	✓	✓	✓
Logos on competition apparel		✓	✓	✓	✓
Free team memorabilia			✓	✓	✓
Ability to showcase at corporate events				✓	✓
Naming team after brand name**					✓
OUR HAPPINESS LEVEL	😊	😊😊	😊😊😊	😊😊😊😊	😊😊😊😊😊😊

Gold and Title packages also allow sponsor guests to join the event alongwith team.

* The amount quoted is the minimum required to qualify for the category.

** The car color will be based on brand theme of sponsor.

PROJECT TIMELINE

2014	July	Aug.	Sept.	Oct.	Nov.	Dec.	Jan.	Feb.
Disassembly of car								
International Order Placement								
Chassis Modification								
Motor Controller Design								
Construction of Fiber Glass body								
Testing and optimization								
Packaging and shipment								
Shell Eco-marathon Asia 2015								



ADVISORY PANEL

PROF. DR. FAZAL AHMAD KHALID, (SITARA-E-IMTIAZ)
C.ENG. D.PHIL. (OXON), MASM, FIM, FRMS & FPAS
PRO-RECTOR GIKI, PROFESSOR, FACULTY OF MATERIAL SCIENCE & ENGINEERING, GIKI.

PROF. DR. JAVED AHMAD CHATTHA
PH.D. (UNIVERSITY OF BIRMINGHAM, UK)
PROFESSOR, FACULTY OF MECHANICAL ENGINEERING, GIKI.

DR. NISAR AHMED
PH.D. (UNIVERSITY OF LONDON, IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE, LONDON, UK)
DEAN AND PROFESSOR, FACULTY OF ELECTRICAL ENGINEERING, GIKI.

DR. NOOR MUHAMMAD
MBA, PH.D. (UNIVERSITY OF HUDDERSFIELD, UK)
ASSISTANT PROFESSOR, FACULTY OF MANAGEMENT SCIENCES AND HUMANITIES, GIKI.



PREVIOUS SPONSORS





www.twitter.com/hammerheadarc



www.facebook.com/hammerheadgiki



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Photography, design, and content
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